

Finite-depth Jacobian growth in GNNs

What random probes can estimate, what adjoints change, and how far forward-only methods reach

2026-09-27

Executive summary

The question

For a trained graph neural network, how should we estimate the largest depth-wise growth of a hidden-state perturbation, and what accuracy is actually attainable with forward-only random probes? This report studies a finite-depth quantity on controlled synthetic graphs, validates competing estimators against exact small-system references, and measures wall-clock cost.

The quantity is finite-depth and unprojected

For a fixed graph G , input x , trained parameters Φ , and hidden-state trajectory $h_0 \mapsto h_K$, define

$$B_K = D_{h_0}(F_{K-1,G} \circ \dots \circ F_{0,G})(h_0) = J_{K-1} \dots J_0, \quad \lambda_K^{\text{top}} = \frac{1}{K} \log \sigma_1(B_K).$$

The derivative includes the complete implemented layer: activation, normalization, residual paths, and state-dependent geometry. It excludes encoder and readout derivatives. No mean-centering or between-layer projection is inserted. Pierre’s projected cocycle is a separate observable, not an estimator of B_K .

This is a finite-depth sensitivity rate. It is not automatically an asymptotic Lyapunov exponent, an end-to-end prediction sensitivity, an oversquashing diagnostic, or a task-utility score.

The main conclusions

1. **Random full-space probes are valid lower bounds but poor point estimators in this pilot.** The maximum over M independent JVP probes is monotone in M and converges almost surely for a fixed finite product, but it remained far from the exact rate after $M = 2^{20}$ on the recent same-cell study: gaps were 0.26–0.35 for GCN/ChebNet and 0.085–0.097 for the sheaf baseline. The earlier broad calibration already found zero target coverage at $M = 1024$ for every reported model/family cell.
2. **If VJPs are available, short operator-refined probes are the practical solution.** Power iteration on $B_K^* B_K$ is an adjoint-enabled refinement of a probe, not a new finite-depth target. On the nine recent operators, eight of nine reached gap $\leq 10^{-4}$ by step 4; six of nine reached $\leq 10^{-12}$ by step 8. Early stopping and Lanczos reduced matrix-free wall-clock by 5–15 $\times$ relative to fixed three-start, 160-step power.
3. **Under a strict JVP-only restriction, graph-smoothed probes were the best tested variant.** Drawing full-space probes from $q \propto (I + cL)^{-1}z$ improved the gap

at $M = 2^{20}$ by about $1.8\text{--}2.1\times$ relative to iid probes at similar wall-clock. A small- M follow-up found that even ten smoothed probes gave pooled-best gaps of 0.17 (GCN), 0.21 (ChebNet), and 0.08 (sheaf); at $M = 2^{10}$ these were about 0.09, 0.09, and 0.04. These are useful screening bounds, not tight estimates.

4. **The forward-only limitation is still a large gap.** The smoothed-probe curves retained the descriptive $A - B \log_2 \log_2 M$ shape; extrapolations for a 0.01 gap were 2^{233} – 2^{258} probes and are explicitly treated as out-of-range fits, not universal theorems. Forward-only B -power was also tested and stalled at eigenvalue/spectral-radius behavior on these nonnormal products rather than converging to $\sigma_1(B_K)$.
5. **Exact rates distinguish the trained baselines, but do not diagnose over-squashing alone.** Across 900 graph/signal draws per checkpoint, the mean exact top rates were GCN 0.804, ChebNet 1.024, sheaf 0.514, and heat diffusion approximately 0. Positive global amplification can coexist with contraction of most singular modes or with poor source-to-target communication.
6. **Pierre’s projection answers a different question.** The literal node-mean projected rate was lower than the full-space rate for every architecture (GCN 0.344 vs 0.674, sheaf 0.427 vs 0.523, ChebNet 0.844 vs 1.112, matched-cell means), but it is the rate of a projected cocycle and must remain labeled separately.

The strongest practical recommendation is therefore conditional: use dense SVD for tiny calibration systems; use residual-checked $B_K^* B_K$ power or Lanczos when adjoints are available; and use graph-smoothed full-space multiprobe only as an explicitly labeled JVP-only lower bound. No universal probe count is supported.

Authoring and execution context

This revision was prepared in the **pi coding-agent harness**, session ID 01a0ceaa-cf63-77a8-8431-38ac76a7a7b5. Jacobian calculations, tests, plotting, and PDF builds were executed on Slurm compute nodes; file inspection and version-control operations were performed on the login node. The final source commit, lockfile hash, job IDs, manifests, and raw-result paths are recorded in RESEARCH_LOG.md.

1. Estimand and estimator families

1.1 The hidden-state product

Let $h_0 = e_G(x)$ be the encoder output and let

$$h_{\ell+1} = F_{\ell,G}(h_\ell), \quad \ell = 0, \dots, K-1.$$

The product is

$$B_K = J_{K-1} \cdots J_0, \quad J_\ell = D_{h_\ell} F_{\ell,G}(h_\ell),$$

and the target is

$$\lambda_K^{\text{top}}(G, x; \Phi) = K^{-1} \log \sigma_{\max}(B_K).$$

All norms are Euclidean/Frobenius norms after flattening the hidden tensor. Graph structure, parameters, evaluation-mode stochastic state, and the forward trajectory are frozen. For state-dependent sheaf layers, the derivative includes the path $h \rightarrow R(h) \rightarrow L_{\mathcal{F}(h)} h$; detaching the geometry changes the model being measured.

The target does not include encoder/readout derivatives. If end-to-end input-output sensitivity is desired, it must be reported separately as a derivative of the complete prediction map. A finite feed-forward network also does not define an asymptotic Lyapunov exponent merely by being written as a sequence of layers; an asymptotic claim needs an extendable cocycle and a specified limiting regime.

1.2 What each estimator returns

Method	Output	Strength	Limitation and use
Dense Jacobian plus SVD	Exact λ_K^{top}	Trusted tiny-system reference and full spectrum	D^2 storage and roughly D directional derivatives; not scalable
Full-space multiprobe JVP	$\max_{m \leq M} K^{-1} \log \ B_K q_m\ $	JVP-only, parallel, monotone lower bound	No finite universal error certificate; can be very loose
Power on $B_K^* B_K$	Approximation to λ_K^{top}	Few operator products; residual directly checkable	Requires VJPs and a correct adjoint through every layer
Matrix-free Lanczos on $B_K^* B_K$	Ritz approximation to the same top singular value	Often fewer iterations; residual available	Requires VJPs and reorthogonalization
Graph-smoothed multiprobe	JVP-only lower bound with graph-informed probe distribution	Best forward-only variant tested here	Distribution is a heuristic; c-scan is limited to this pilot
Forward-only B -power	Growth after repeatedly applying B_K	Cheap JVP-only refinement attempt	Targets eigenvalue behavior for nonnormal products, not generally $\sigma_1(B_K)$

1.3 The two important distinctions

First, a finite- M multiprobe estimate is a lower bound:

$$\hat{\lambda}_{K,M} = \max_{m \leq M} \frac{1}{K} \log \|B_K q_m\| \leq \lambda_K^{\text{top}}.$$

For a nested iid stream it is nondecreasing in M and converges almost surely for each fixed finite product. It is not unbiased and does not certify a small gap at finite M .

Second, a power step is not a forward-only probe step. Applying $B_K^* B_K$ requires a forward JVP sweep and a reverse VJP sweep. That extra adjoint is exactly what lets the algorithm seek the dominant right-singular direction rather than merely wait for a random probe to land near it.

2. Study design and evidence tiers

2.1 Architectures

The controlled baselines are intentionally small diagnostic models:

- **GCN:** depth 3, $h_{\ell+1} = \text{ReLU}(\tilde{A} h_\ell W_\ell)$, with self-looped symmetric normalized adjacency.
- **ChebNet:** depth 3, order-3 Chebyshev filtering of the self-loop-free symmetric normalized Laplacian, with the configured degree-3 polynomial contrast.
- **Feature-conditioned diagonal-sheaf baseline:** depth 10, endpoint-dependent diagonal restriction maps, assembled sheaf-Laplacian action, residual update and ReLU. It is inspired by Neural Sheaf Diffusion but is not claimed to reproduce every detail of that paper; its implemented normalization is one-sided row normalization.

All models use four-dimensional inputs and hidden width 12 in the retained checkpoints. Exact equations and implementation choices are in `notes/experiment-baselines.md`.

2.2 Data and training

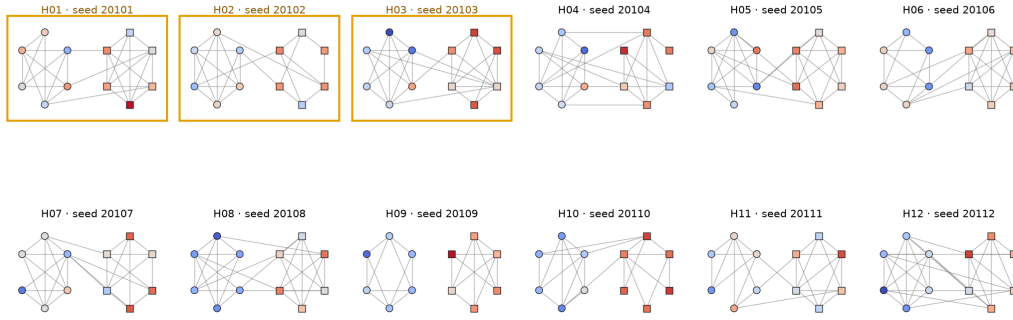
The supervised pilot uses deterministic synthetic node-classification data. Each node receives $(2y_i - 1)e_1 + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, I_4)$. The three graph conditions are:

- a 12-node homophilic contextual SBM, $p_{\text{same}} = 0.75$, $p_{\text{other}} = 0.10$;
- a 12-node heterophilic SBM with those probabilities swapped;
- a 15-node barbell with two six-node cliques and a three-edge bridge.

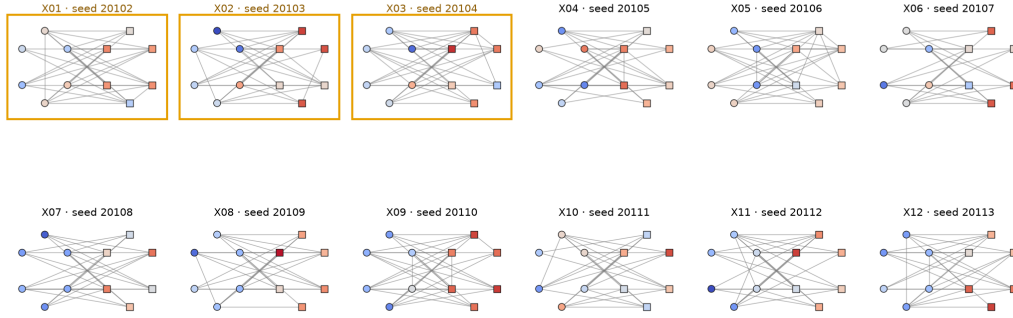
The split is 16 training graphs, 6 validation graphs, and 36 test graphs: 12 test graphs in each condition. Training uses 120 Adam epochs, weight decay 10^{-4} , validation selection over widths $\{8, 12\}$ and learning rates $\{0.003, 0.01\}$, and two model seeds. The held-out homophilic accuracy means were GCN 0.865, ChebNet 0.861, and sheaf 0.878; heterophilic accuracy was 0.312, 0.524, and 0.469 respectively. These figures establish that the checkpoints are trained task models, not a claim of broad architecture superiority.

All 36 held-out test graph/signal examples

Homophilic SBM



Heterophilic SBM



Barbell

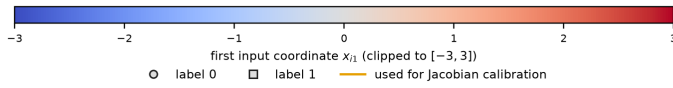
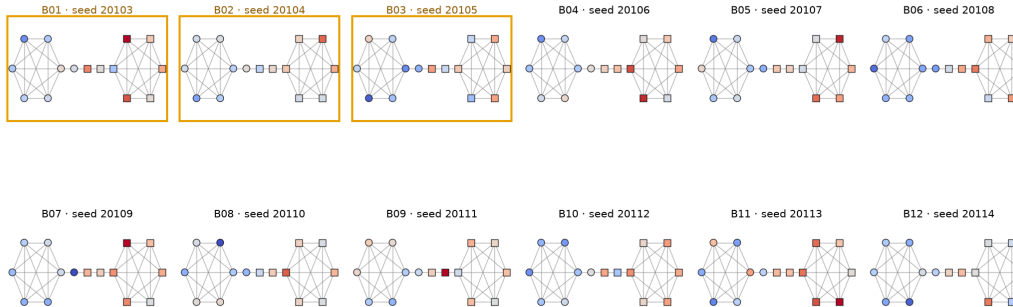


Figure 1. The 36 held-out graph/signal examples. Gold frames mark the nine unique calibration examples used for the main small-cell Jacobian study.

2.3 Evidence tiers and sample counts

The report combines several experiments; they should not be pooled as if they had the same scope.

Evidence tier	Operators / draws	What it establishes
Broad calibration	54 trained cells: 18 per architecture, from two model seeds, three families, and three held-out trajectories per family	Initial probe, dense-reference, and power calibration
Recent probe sweep	9 operators: one seed-90000 example per family and one checkpoint per architecture; two independent streams per operator	$M = 2^{10} \dots 2^{20}$ convergence and same-cell method comparisons
Recent forward-only study	The same nine operators; iid and graph-smoothed streams, plus small- M follow-up	Wall-clock Pareto and tested JVP-only alternatives
Histograms	900 (G, x) draws per one trained checkpoint/model	Descriptive rate distributions and leading singular-mode distributions
Pierre comparison	54 matched projected cells	Separate projected-cocycle observable

The recent nine-operator studies are deliberately narrower than the broad calibration. Their value is controlled comparison, not architecture-wide generalization.

2.4 Validation and reproducibility

The implementation has tests for explicit-vs-JVP products, zero products, unit-probe enforcement, a GCN constant-mode/no-projection check, power-vs-dense linear products, and non-linear GCN/sheaf JVP products against independent reverse-mode dense autodiff. The final Slurm validation at commit 63cd2e4 passed 24 tests. Float64 is used for calibration and comparison. The full lockfile hash, environment, seeds, checkpoints, graph conditions, job IDs, raw paths and artifact checksums are in each tracked manifest and RESEARCH_LOG.md.

3. What random probes actually achieve

3.1 The broad calibration: $M \leq 1024$ is not a point-estimation budget

The initial nested-stream calibration used the full tangent space, no projection, independent streams, and exact dense-SVD references on tiny cells. At $M = 1024$, target coverage at $\text{gap} \leq 0.01$ was 0/72 for every reported model/family cell. Pooled median gaps were approximately 0.395 for GCN and 0.109 for sheaf; the independent ChebNet replication gave medians 0.347, 0.346, and 0.376 for homophilic, heterophilic, and barbell families.

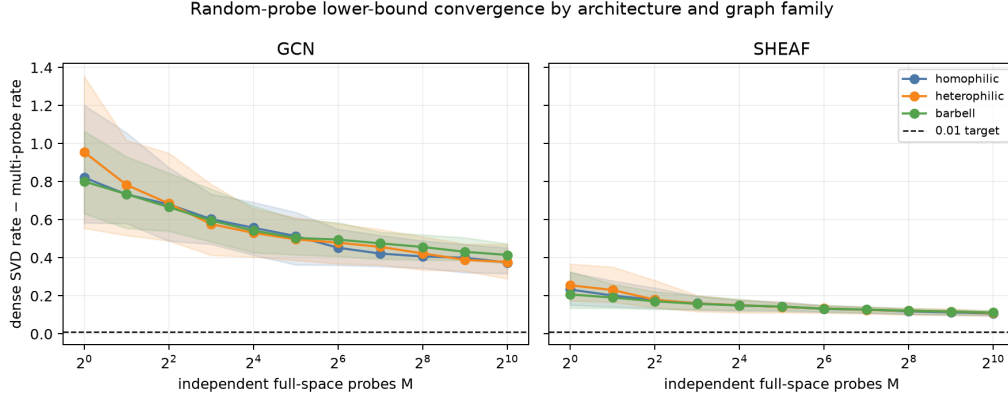


Figure 2. The initial $M \leq 1024$ calibration. The maximum is a valid lower bound, but it is not close enough to the exact rate to be used as an unqualified point estimate in these cells.

3.2 The extended sweep: one million probes still leaves a visible gap

The extended run streamed iid full-space probes in chunks of 4096 to $M = 2^{20} = 1,048,576$. It completed all 18 streams without cutoff. One stream took roughly 9-11 seconds for GCN, 15-19 seconds for ChebNet, and 263-329 seconds for the depth-10 sheaf. Median gaps decreased only modestly from $M = 2^{10}$ to 2^{20} :

Model	$M = 2^{10}$	$M = 2^{16}$	$M = 2^{20}$	Relative reduction
GCN	0.390	0.355	0.318	~18% per 1024×
ChebNet	0.380	0.344	0.308	~19% per 1024×
Sheaf	0.117	0.105	0.090	~23% per 1024×

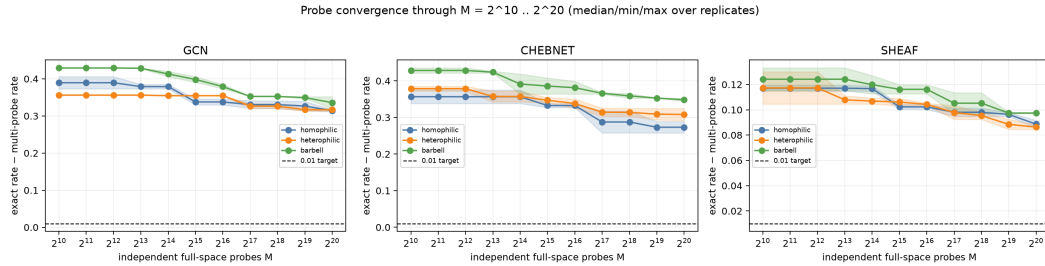


Figure 3. The extended iid-probe sweep. The monotone lower bound improves, but the target gap is not approached at a feasible probe count.

A fit of the form $A - B \log_2 \log_2 M$ describes the measured range. Its extreme extrapolation should be treated as descriptive only: depending on the operator and distribution, the fitted M for a 0.01 gap lies around 2^{245} or larger. This is evidence of an impractical overlap-concentration regime, not a universal theorem about all GNN Jacobians.

4. What adjoints change: operator-refined probes

4.1 Short $B_K^* B_K$ refinement

The same nine recent operators were used to compare random probes with an operator-refined probe. Starting from a random unit vector, each refinement applies

$$v \leftarrow \frac{B_K^* B_K v}{\|B_K^* B_K v\|}.$$

The rate is then evaluated from $K^{-1} \log \|B_K v\|$. This is ordinary power iteration on the symmetric positive semidefinite operator $B_K^* B_K$, expressed in the language of adaptive probes.

A probe-estimator variation that converges way faster: refine one probe instead of drawing 10^6

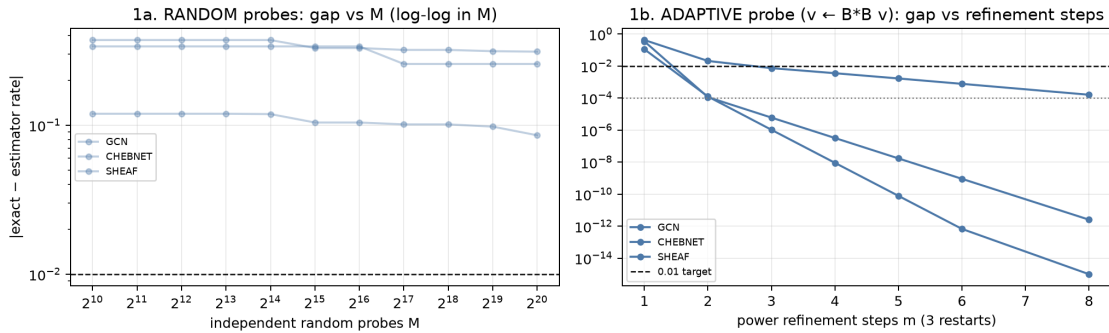


Figure 4. On identical operators, random probes improve slowly with M , while adjoint-enabled refinement reaches the reference in a handful of steps.

Across the nine operators, 8/9 reached rate gap $\leq 10^{-4}$ by step 4; 6/9 reached $\leq 10^{-12}$ by step 8. The slow outlier was a ChebNet/homophilic operator whose gap remained about 1.6×10^{-4} at step 8. Thus the safe statement is “most cells reached 10^{-4} in 2-4 steps,” not that every cell did.

Model	Random gap at 2^{20}	Adaptive gap at step 2	Adaptive gap at step 3	Adaptive gap at step 6
GCN	0.31-0.35	10^{-5} - 1.3×10^{-3}	6×10^{-6} - 3.1×10^{-5}	about 10^{-12}
ChebNet	0.26-0.34	1.3×10^{-2} - 2.1×10^{-2}	6×10^{-6} - 7.4×10^{-3}	1.7×10^{-3} - 10^{-11}
Sheaf	0.085-0.097	2.4×10^{-5} - 1.8×10^{-4}	5×10^{-8} - 10^{-6}	about 10^{-14}

Each refinement costs one forward JVP sweep and one reverse VJP sweep. This is why the convergence improvement is so large: the method actively aligns with the right-singular direction rather than waiting for a random draw.

4.2 Matrix-free speedups and residual gates

For the same nine operators, early-stopping power and Lanczos were compared with fixed three-start, 160-step power. Median seconds per unique operator, with three timed repeats, were:

Model	Fixed power	Early power	Lanczos	Early speedup	Lanczos speedup
GCN	1.26	0.086	0.12	$14.7\times$	$10.4\times$
ChebNet	2.64	0.36	0.49	$7.3\times$	$5.4\times$
Sheaf	30.7	2.06	5.9	$14.9\times$	$5.2\times$

The fixed benchmark has three unique operators per model and three timed repeats per operator. All timed early-power and Lanczos calls passed their residual gate and matched the exact small-system reference to numerical precision. At these tiny tangent dimensions, dense SVD is still faster; the matrix-free result matters for the dimension regime where dense storage is impossible.

A fine-grained stopping experiment also compared residual thresholds 10^{-7} and 10^{-2} . With checks every 10 sweeps, the thresholds often stopped at the same checkpoint. Checking every sweep let the looser threshold stop earlier: median times were 0.045, 0.136, and 1.09 seconds for GCN, ChebNet, and sheaf, with maximum observed rate gaps 3.1×10^{-5} , 2.5×10^{-4} , and 2.4×10^{-5} in that timing sample. This is a calibrated speed/accuracy option, not a universal residual-to-rate error theorem.

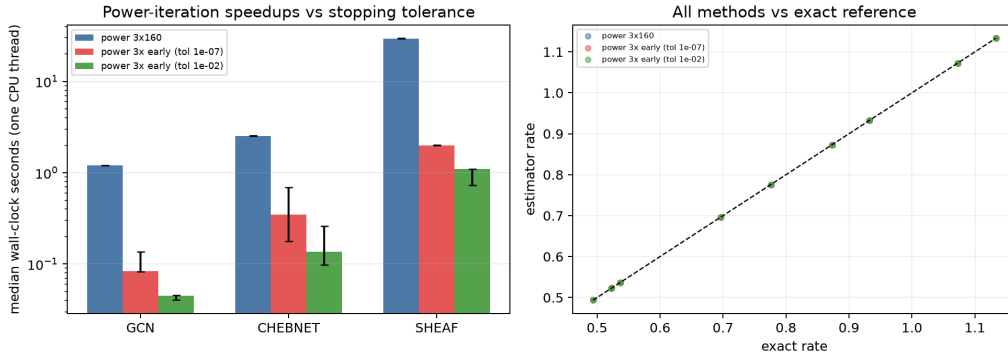


Figure 5. The fixed, early-stopped, loose-threshold, and Lanczos timing comparison.

5. The best tested JVP-only method

5.1 Graph-smoothed full-space probes

The strict forward-only setting permits JVP applications of B_K but no adjoint. In that setting, forward B -power is not a valid replacement for singular-value iteration on general nonnormal products: its direction is eigenvalue-driven and empirically stalled below the top singular rate by gaps of 0.11–0.57 after 40 steps on these cells.

The remaining tested lever is the probe distribution. We drew full-space probes using

$$q = \frac{(I + cL)^{-1}z}{\|(I + cL)^{-1}z\|},$$

where L is the normalized graph Laplacian and z is Gaussian. For finite c , this remains a full-support distribution and therefore preserves the same asymptotic target in principle; it is a graph-informed importance distribution, not a projection.

On the same nine recent operators, $c = 1$ smoothed probes dominated iid probes at every recorded marker. At $M = 2^{20}$:

Model	iid gap	Smoothed gap	Median wall-clock at 2^{20}
GCN	0.313	0.166	8.7 / 9.8 s
ChebNet	0.292	0.154	15.0 / 16.2 s
Sheaf	0.086	0.045	280.5 / 280.9 s

The c -scan was performed on one homophilic operator per architecture, at $M = 2^{14}$, and is therefore a local sensitivity check rather than a universal tuning rule:

c	GCN	ChebNet	Sheaf
0 (iid)	0.373	0.338	0.119
0.5	0.262	0.245	0.090
1	0.205	0.197	0.076
4	0.109	0.114	0.051
16	0.091	0.077	0.039

Best forward-only estimator wall-clock Pareto (forward probes only; rightmost point = $M = 2^{20}$)

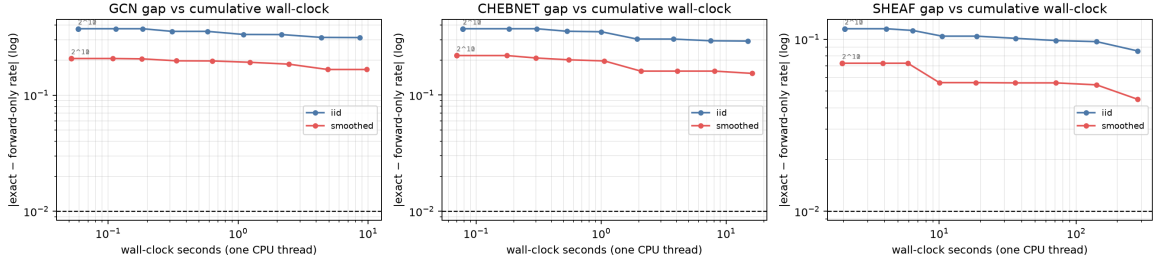


Figure 6. Gap versus cumulative wall-clock for iid and graph-smoothed forward-only probes. Probe work is batched; the total wall-clock still grows with M , especially for the deep sheaf.

The marginal batched probe costs were approximately 8 microseconds for GCN, 14 microseconds for ChebNet, and 0.25 milliseconds for sheaf in this CPU benchmark. The total cost is not flat: reaching 2^{20} cost roughly 9–10 seconds for GCN, 15–16 seconds for ChebNet, and 281 seconds for sheaf. The important limitation is that the gap decreases too slowly for this cost to buy a 0.01 estimate.

5.2 Very small probe counts

A follow-up measured $M \in \{10, 32, 128, 512, 1024, 4096, 16384\}$ with five independent streams on one homophilic operator per architecture. The table reports the median gap across streams and, in parentheses, the best pooled gap among those five streams; the pooled value is optimistic and should not be confused with a typical run.

Model, smoothed $c = 16$	$M = 10$	$M = 2^{10}$	$M = 2^{12}$	$M = 2^{14}$
GCN	0.30 (0.17)	0.15 (0.09)	0.12 (0.09)	0.09 (0.08)
ChebNet	0.28 (0.21)	0.13 (0.09)	0.11 (0.09)	0.09 (0.08)
Sheaf	0.09 (0.08)	0.045 (0.04)	0.042 (0.03)	0.039 (0.03)

For comparison, the GCN iid median gaps were 0.61, 0.41, and 0.37 at $M = 10$, 2^{10} , and 2^{14} respectively. A frequency-adaptive reweighting of probes toward nodes used by the best probes was also tested. It spent probes learning the reweighting and was worse than fixed smoothing at every tested count; for example, GCN at $M = 10$ was 0.37 adapted

versus 0.17 for the pooled-best smoothed stream. Fixed graph smoothing is therefore the best JVP-only distribution tested here, not a proof of a universal optimum.

5.3 The practical JVP-only conclusion

The forward-only result is deliberately modest:

- graph-smoothed probes are a real, reproducible improvement over iid probes on these synthetic operators;
- ten probes can give a useful coarse lower bound, especially for the sheaf baseline, but not a tight top-rate estimate;
- at $M = 2^{10}$, the best tested pooled gaps remain about 0.04–0.09 for the displayed operators;
- the fitted double-log behavior means that a 0.01 target is not feasible in this study by forward-only probing alone.

This should be reported as a **lower-bound screening method**. If the scientific question requires a tight top singular rate, the correct escape is an adjoint-enabled method, not more confidence in a large finite- M maximum.

6. What the exact rates reveal about these models

6.1 Rate distributions

The histogram diagnostic loads checkpoints and never trains. It draws the same 900 graph/signal pairs per trained checkpoint: 300 per family. The exact reverse-mode dense Jacobian/SVD rates have means:

System	Mean	Median	Standard deviation	Homophilic	Heterophilic	Barbell
GCN	0.804	0.793	0.068	0.792	0.825	0.796
ChebNet	1.024	1.028	0.076	0.996	1.091	0.984
Sheaf	0.514	0.517	0.026	0.506	0.528	0.507
Heat diffusion	approximately 0	approximately 0	approximately 10^{-17}	approximately 0	approximately 0	approximately 0

Exact top (1st) eigenmode rates over random (G, x) pairs (same 300 draws/family)

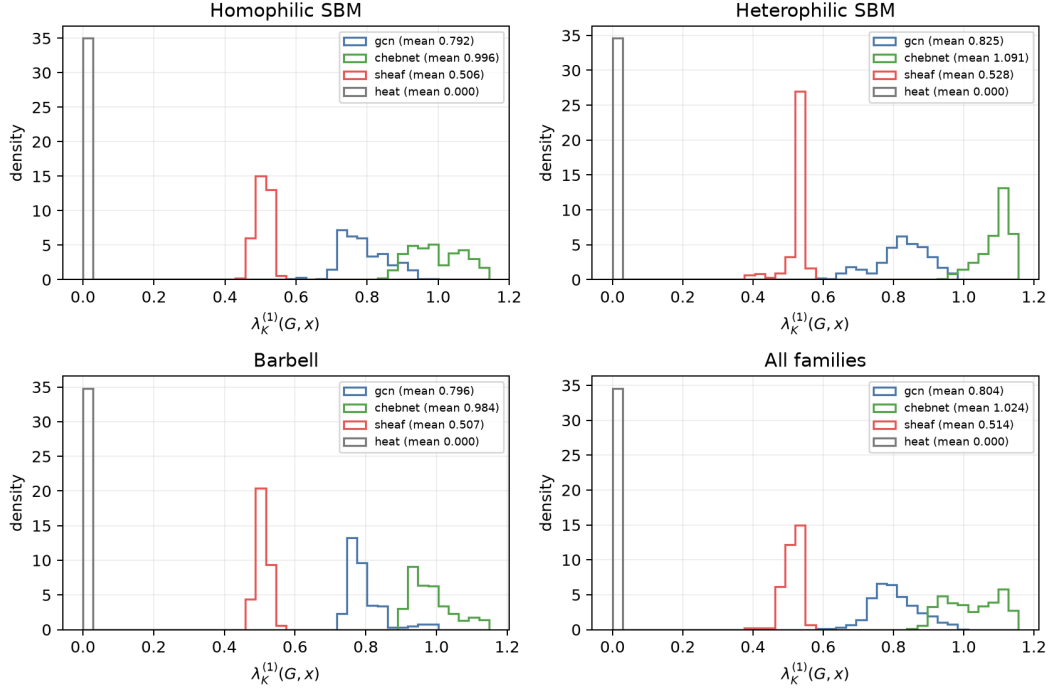


Figure 7. Exact finite-depth top-rate distributions for one trained checkpoint per GNN and for the heat control. These are descriptive draws, not population-level architecture claims.

ChebNet has the largest mean top rate, followed by GCN and the sheaf baseline. The distributions are narrow on these synthetic draws, but one checkpoint per architecture is not enough to establish a general architecture ordering. Heat diffusion provides a non-expanding control: with the symmetric normalized Laplacian, the preserved mode is a normalized harmonic mode, not necessarily the raw constant vector; the remaining modes contract.

6.2 Leading singular modes

The exact diagnostic also computed the leading 25 singular values per draw. The following table uses modes 1, 13, and 25 for GNNs and modes 1, 2, and 3 for heat:

System	Mode 1	Mode 13 / 2	Mode 25 / 3	Interpretation
GCN	0.804	-0.733	-1.473	Growth concentrated in a few directions Positive mean persists through reported mode 25 Gradual spectral decay
ChebNet	1.024	0.389	0.248	
Sheaf	0.514	0.125	0.040	

System	Mode 1	Mode 13 / 2	Mode 25 / 3	Interpretation
Heat	approximately 0	-0.072	-0.155	Harmonic mode preserved; other modes contract

Eigenmode rate histograms per system (GNNs: modes 1/13/25; heat: modes 1/2/3)

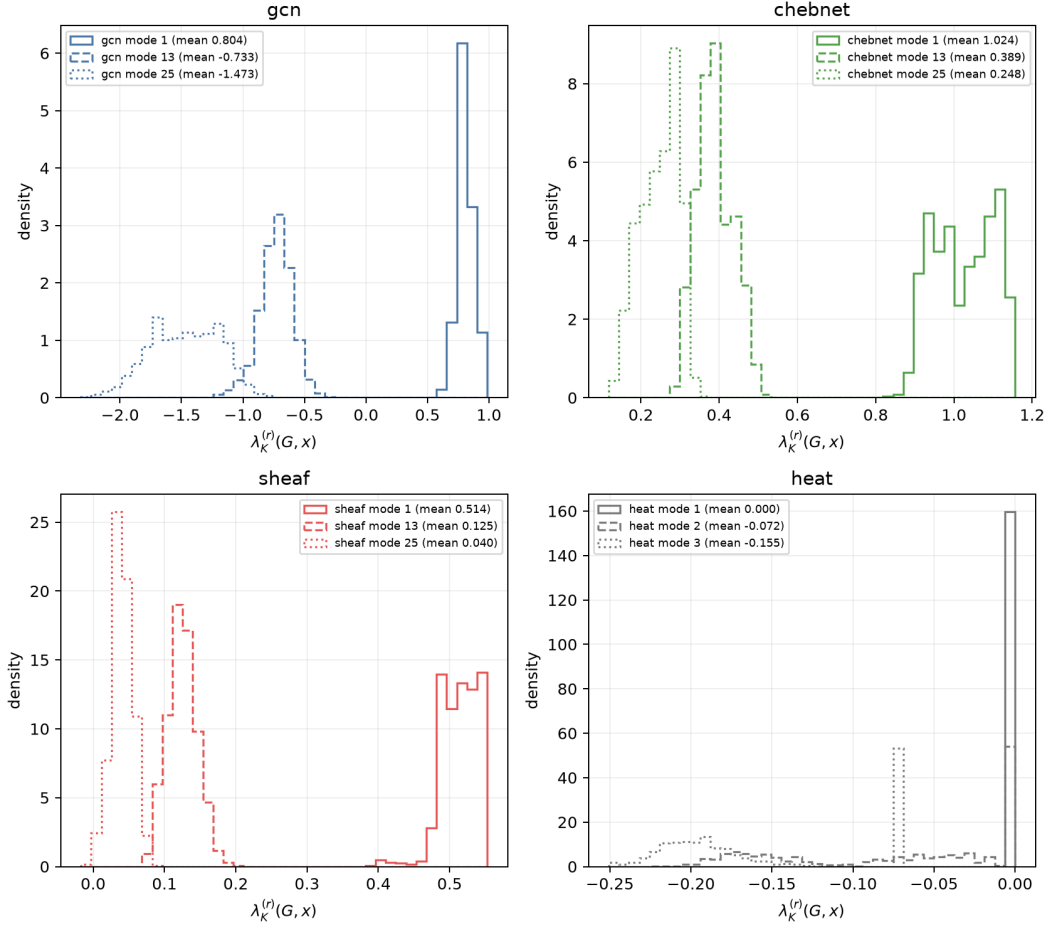


Figure 8. These are singular-mode rates, not eigenvalue rates. “Positive through mode 25” refers to the reported mean, not every individual draw.

A positive global top rate is compatible with contraction of most directions. It also says nothing by itself about individual source-to-target Jacobian blocks, delocalization, oversquashing, or classification performance. Those require separate observables.

7. Pierre’s projected observable

Pierre’s literal projected cocycle initializes probes in the node-mean-zero subspace and applies

$$B_K^P = PJ_{K-1}PJ_{K-2}\cdots PJ_0|_{\mathbf{1}_N^\perp}, \quad Pz = z - \frac{1}{N}\mathbf{1}\mathbf{1}^\top z$$

after every layer. This can be a meaningful transverse observable, but

$$B_K^P \neq B_K$$

in general. For normalized graph Laplacians the natural harmonic vector is degree-weighted, and for sheaves the harmonic space depends on the learned restriction maps; the literal node-mean prescription is a deliberate choice, not an invariant-subspace theorem.

The matched projected calibration used 54 cells per estimator family, 12 nested streams through $M = 1024$, and projected dense references. Mean exact rates were:

Model	Full-space B_K	Projected B_K^P	Difference
GCN	0.674	0.344	-0.330
Sheaf	0.523	0.427	-0.096
ChebNet	1.112	0.844	-0.268

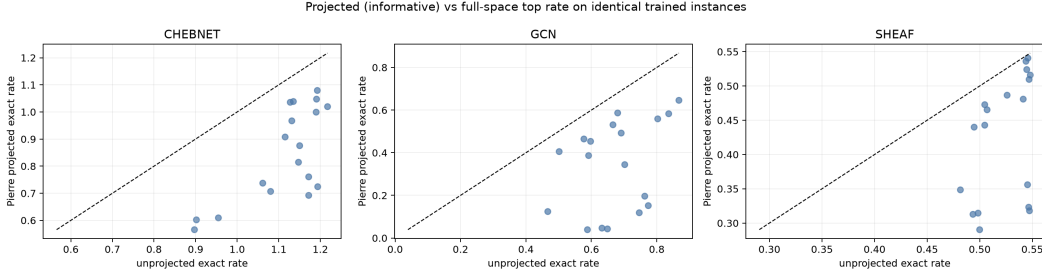


Figure 9. Projection lowers the measured top rate on these matched cells, but the projected quantity remains positive on average. The comparison is between two estimands, not two implementations of one estimator.

At $M = 1024$, projected probes also missed their exact target. Their finite- M averages were GCN -0.038, sheaf 0.324, and ChebNet 0.517, versus full-space values 0.280, 0.414, and 0.759. These are lower bounds, not substitutes for the projected dense reference.

8. Recommended protocol

Situation	Recommended estimator	Acceptance rule and reporting
Tiny calibration system	Dense reverse-mode Jacobian/SVD, independently checked against JVP product	Report D , product discrepancy, singular spectrum, state boundary and precision
Tight finite-depth top rate with VJPs	Multi-start power on $B_K^* B_K$ with residual checks; use Lanczos if convergence is slow	Report each start, residual, iterations, JVP/VJP counts, accepted estimate and exact-SVD comparison where available
JVP-only screening	Batched full-space graph-smoothed multiprobe	Report distribution, c , M , independent streams, wall-clock, and the nonnegative gap interpretation; never call it an exact exponent
Unknown or large system	Calibrate on a smaller matched instance first	Do not transfer a probe count or residual threshold without evidence
Long stationary/extendable cocycle	QR/Benettin or another asymptotic-spectrum method	State the stationarity/extendability assumptions; do not identify a finite-network top singular rate with the asymptotic exponent
Oversmoothing/oversquashing question	Separate transverse, blockwise, or source-to-target observables	Global $\sigma_1(B_K)$ is insufficient

The operational defaults are therefore: dense reference when feasible; residual-checked matrix-free power/Lanczos when adjoints exist; graph-smoothed multiprobe as a JVP-only lower-bound screen. No projection P is inserted into the primary target.

9. Limitations and next experiments

- The graphs are small synthetic graphs selected for exact calibration. The recent nine-operator follow-ups and the c -scan are narrower than the broad calibration and do not establish a universal probe distribution or probe count.
- The sheaf model is a diagnostic implementation, not a claim to reproduce every Neural Sheaf Diffusion variant. The state-dependent- μ architecture remains underspecified in notes/state-dependent-mu-diffusion-GNN.md and is not silently treated as implemented.
- Histograms use one trained checkpoint per model. Their model ordering and narrowness are descriptive for these draws, not architecture-level conclusions.
- Each recent (G, x) study varies graph and feature signal jointly. It does not estimate input-only variability on a fixed graph.
- The fitted $A - B \log_2 \log_2 M$ curves are extrapolations beyond the measured range. They motivate caution but are not universal lower bounds.
- A positive global top rate does not prove long-range communication, delocalization, or absence of oversquashing. Future work should measure source-to-target Jacobian blocks, mode localization, and task-relevant subspaces separately.

- The next computational scale-up should keep the exact-SVD calibration suite, then test residual-gated power/Lanczos and forward-only graph-smoothed probes on larger graphs, real datasets, more checkpoints, and explicitly specified adaptive-geometry models.

10. Evidence map and reproducibility

Tracked evidence is under `evidence/<run-id>/`; raw outputs remain under `ignored results/<run-id>/` on the cluster. The most relevant artifacts are:

- 2026-09-24-probe-calibration-v2: broad $M \leq 1024$ calibration and exact references, Slurm 22351.
- 2026-09-24-estimator-wall-clock: dense/probe/fixed-power timing, Slurm 22388.
- 2026-09-24-lyapunov-histogram-comparison: exact 900-draw model/heat histograms and leading singular modes, Slurm 22400/22406 and curation 22402/22407.
- 2026-09-24-pierre-estimator-calibration: matched projected/unprojected study, Slurm 22377/22380.
- 2026-09-26-probe-convergence-2p20: iid stream through 2^{20} , Slurm 22803.
- 2026-09-26-power-speedup: early stopping and Lanczos, Slurm 22796/22803.
- 2026-09-27-probe-vs-adaptive: random versus adjoint-enabled refinement, Slurm 22828.
- 2026-09-27-forward-only-pareto: iid versus graph-smoothed JVP-only probes, Slurm 22846.
- 2026-09-27-forward-adapted: small- M and frequency-adaptive negative result, Slurm 22855.

All experimental conclusions in this report point to tracked figures/tables and manifests. The complete command lines, commits, lockfile hash, Python/PyTorch versions, seeds, checkpoint paths, reference tolerances, failure modes, and Slurm logs are recorded in `RESEARCH_LOG.md`.

References

Local-source provenance is indexed in `papers/README.md`. The core references are:

- Kipf & Welling, Semi-Supervised Classification with Graph Convolutional Networks, arXiv:1609.02907.
- Defferrard, Bresson & Vandergheynst, Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering, arXiv:1606.09375.
- Bodnar et al., Neural Sheaf Diffusion: A Topological Perspective on Heterophily and Oversmoothing in GNNs, arXiv:2202.04579.
- Oono & Suzuki, Graph Neural Networks Exponentially Lose Expressive Power for Node Classification, arXiv:1905.10947.
- Alon & Yahav, On the Bottleneck of Graph Neural Networks and its Practical Implications, arXiv:2006.05205.
- Johansson, Strannegård, Engsner & Mostad, Exact Spectral Norm Regularization for Neural Networks, arXiv:2206.13581.
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- Benettin et al., Lyapunov Characteristic Exponents for Smooth Dynamical Systems, Meccanica 15, 1980.